

# Benjamin Pachter

Principal AI & Data Architect • Full-Stack LLM Engineer

980-297-8331

[ben.pachter@bellsouth.net](mailto:ben.pachter@bellsouth.net)

[linkedin.com/in/benjamin-pachter](https://linkedin.com/in/benjamin-pachter)

Research Triangle Park, NC

## 01 SUMMARY

Principal-level AI & data architect who ships production ML/LLM platforms end to end — data architecture and pipelines, model evaluation, and scalable serving (RAG / vector search, multi-model routing, GPU inference). At Duke Energy, architected a geospatial capacity-modeling platform screening **250+ GW** of transmission load, supported **150+ hyperscaler data-center site analyses**, and cut load-study turnaround from 2–4 weeks to same-day (~200 engineering hours saved / month). Owns platform roadmap and technical standards across five internal organizations, translating ambiguous business needs into production-grade data / AI systems. Full-stack application developer by hobby.

AI / ML Architecture

LLMOps / MLOps

Data Architecture

ETL / ELT

APIs & Microservices

AWS

Docker / Kubernetes

CI / CD

Agentic AI & RAG

Evals & Observability

## 02 TECHNICAL SKILLS

|                        |                                                                                                                                                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agents & LLMOps        | LangGraph, LangChain, CrewAI, MCP (Model Context Protocol), multi-agent orchestration, tool-calling loops, structured outputs, context engineering, DSPy                                                                          |
| RAG & Knowledge Graphs | Grounded RAG with citations, agentic RAG, hybrid search, reranking, GraphRAG; knowledge-graph construction, ontology design (IEC CIM 61970 / 61968), LLM entity / relationship extraction; embeddings, ChromaDB, pgvector, Qdrant |
| Evals & Observability  | Evals & LLM-as-judge (local + cloud), RAGAS, LangSmith, tracing (Langfuse, Phoenix, OpenTelemetry / X-Ray), Sentry, guardrails, prompt-injection defense (OWASP LLM Top 10)                                                       |
| Serving & Tuning       | vLLM, SGLang, Ollama (local RTX 4090), quantization (GGUF / AWQ), prompt caching, multi-model routing & fallback; fine-tuning — LoRA / QLoRA, DPO, SFT, RLHF, Hugging Face                                                        |
| Foundation Models      | Claude (Anthropic SDK), GPT-5.x, Gemini, Gemma, Llama, Qwen, DeepSeek, Mistral                                                                                                                                                    |
| ML / Deep Learning     | PyTorch, TensorFlow, Transformers, XGBoost / LightGBM, CNNs, time-series forecasting, feature engineering; experiment tracking (MLflow, W&B)                                                                                      |
| Backend & APIs         | Python (FastAPI, Django), Node.js, Express, TypeScript, C# / .NET 8 (ASP.NET Core, xUnit), REST & SSE streaming, WebSockets, microservices; JWT / RBAC auth, 2FA (TOTP), rate limiting, multi-tenant SaaS                         |
| Data Engineering       | SQL, Pandas, NumPy, PySpark, Airflow, ETL / ELT, multi-source connector ingestion, scheduled pipelines (cron), canonical schema design; Java, C++, Scala, Bash                                                                    |
| Cloud & Infra          | AWS (S3, Lambda, Glue, Athena, Redshift, SageMaker, Bedrock, App Runner, CloudFront, CDK, OIDC CI), Docker, Kubernetes, CI/CD, Railway, Cloudflare R2, PostgreSQL, SQLite, CUDA                                                   |
| Geospatial             | ArcGIS Pro, Python toolboxes, spatial raster modeling, IDW interpolation, transmission capacity analysis, PSS/E integration, IEC CIM / CGMES                                                                                      |
| Frontend & Dataviz     | React, TypeScript, Vite, MUI, Emotion, TanStack Query; Three.js / react-three-fiber, D3 (force-directed graphs), Plotly, MapLibre GL, lightweight-charts, Recharts                                                                |

## 03 EXPERIENCE

### Duke Energy

Sep 2024 — Present

#### Senior Business Analyst

Charlotte, NC • Remote

- Architected and delivered a spatial transmission-capacity modeling platform (ArcGIS Pro + Python toolbox) generating a continuous MW-headroom raster (IDW interpolation on tower-level TrLim); screens **250+ GW** of large-load requests and has supported **150+ hyperscaler** data-center site analyses across a 6-state territory.
- Designed CIM-aligned, AI-ready data architecture making grid and asset data interoperable for enterprise-wide analytics, ML, and LLM applications.

- Built prompt-driven and agentic workflows (LangChain) automating site screening, feasibility narratives, and decision support — pairing structured enterprise data with LLM reasoning to cut manual analysis loops.
- Reduced PSS/E load-hosting study turnaround from **2–4 weeks to same-day** by pre-screening out infeasible sites before engineering analysis; saved **~200 engineering hours / month**.
- Engineered production pipelines across AWS, SAP S/4HANA, Salesforce (Apex / Flows), and ArcGIS for 5 internal organizations; automated operational reporting and a major industrial customer's monthly billing.
- Own platform roadmap, technical standards, and stakeholder alignment across transmission planning, economic development, and operations.

## New York Mortgage Trust

Jan 2022 — May 2024

### Acquisitions Analyst

Charlotte, NC

- Built multi-source ingestion pipelines (Python, SQL) integrating Yardi and public-data APIs (FRED, FDIC, BLS, Census); delivered TensorFlow / XGBoost forecasting models for rent, occupancy, and cap rate to support institutional underwriting.
- Deployed CNN document parsing and BERT / Transformer NLP pipelines for document understanding and sentiment analysis; migrated legacy workflows to cloud-native AWS / Azure services.
- Delivered ML-driven risk scoring and executive analytics across a **\$7B+** residential mortgage portfolio; implemented Salesforce automations (Apex triggers, Flow) to streamline operations.

## Deloitte

Jan 2020 — Jan 2022

### Risk & Financial Advisory Analyst

Seattle, WA

- Designed automated ETL pipelines (Python, AWS) and ML models (Random Forest, GBM, ARIMA, GARCH, LSTM) for portfolio risk and bond-volatility forecasting across finance, energy, and defense clients.
- Built NLP pipelines for financial sentiment analysis across news and social media; supported SAP S/4HANA + Azure SQL migrations and delivered audit-automation frameworks for Fortune 500 clients.

## BMW Manufacturing

Jan 2018 — Dec 2018

### Supply Chain Planning Co-op

Greer, SC

- Drove **~\$13M** annual transportation-cost reduction by moving the reorder reserve off every production truck onto a dedicated low-volume milk-run, freeing all 20 rack slots per truck; built time-series forecasting models and Power BI dashboards for supply-chain disruption risk across G05–G09 vehicle platforms.

## 04 SELECTED PROJECTS

### Thessa

thessa.space

#### Founder & Lead Engineer

Open-source (MIT) intelligence terminal over a supply-chain knowledge graph — who depends on whom across space, defense, AI infrastructure, energy, and critical minerals — from primary public records only.

- **Provenance-first graph** — **41,330** typed nodes across 10 types (15,184 organizations) and **52,011** typed edges across 20 relation kinds, plus **78,226** federal awards worth **\$1.98T**, 97.5% of it resolving to a recipient already in the graph.
- **Trust tiers, not a confidence score** — a stored confidence number measured *inverting* against audited accuracy, so tiers are now derived at read time from source class and audit verdict; 74.9% of governed edges sit at verified-or-better, and refuted claims stay in an audit ledger rather than disappearing.
- **Hybrid model fleet** — role-specialized open-weight agents (Qwen, Gemma, Llama, nomic-embed, faster-Whisper) on a 24 GB RTX 4090 run research, extraction, embedding and first-pass adjudication at ~\$0; cloud Claude drives the analyst UI and escalated verification. Local-first, cloud-as-fallback, budget-capped.

Stack — React · TypeScript · Node / Express · SQLite · Anthropic SDK · Ollama · Docker · Railway · R2 · 15+ connectors

## Astraea

[github.com/bpachter/astraea](https://github.com/bpachter/astraea)

### Governance plane for AI-augmented curation

AI proposes facts, deterministic code verifies them, a human publishes. Extracted from Thessa; runs on AWS for ~\$6–9 / month.

- **Named refusal, never "failed"** — a typed C# / .NET 8 reconciliation gate returns **twelve named failure modes**, served as data from a versioned endpoint so one vocabulary drives review-queue filters and metric dimensions across four runtimes.
- **Structurally impossible to bypass** — the Django adjudication portal exposes no update endpoint, returns **HTTP 409** unless the verdict passes, and renders every gate field read-only; on its first run it refused type demotions on nodes holding **\$10.9B** across 632 federal contracts that a name-shape heuristic had misread.
- **Calibrated, not asserted** — split-conformal calibration maps raw model confidence to measured precision over **1,691** human adjudications; with no fitted table the API returns 404, never a guess.

Stack — C# / .NET 8 · ASP.NET Core · xUnit · Django 5 / DRF · React · three.js · AWS App Runner · CDK · OIDC-only CI · OpenTelemetry

## Headroom

[bpachter.github.io/headroom](https://bpachter.github.io/headroom)

Playable transmission-interconnection screen — drop a data center on a 14-bus grid and watch the answer flip from energize to queued, with the binding constraint named. Real screening math: B- $\theta$  DC power flow, PTDF sensitivities, N-1 contingency sweep, islanding detection, validated against hand-solved cases in a zero-dependency test suite. The public, synthetic cousin of the production platform at Duke.

Stack — Vanilla JS · Canvas · zero dependencies · unit-tested engine

## Open-Weight Agent Safety

[github.com/bpachter/open-weight-agent-safety](https://github.com/bpachter/open-weight-agent-safety)

Measurement research on indirect prompt injection against locally-hosted open-weight agents. Attack success decomposes into delivery × obedience × execution rather than one rate, exposing models that score "safe" on naive benchmarks only because they never called the carrier tool — one recorded 0 successes in 258 trials without ever being attacked. Three-armed controls separate effect from base rate, on one GPU.

Stack — Python · Ollama · quantized open weights · factorial design with replication

## Nullius

[bpachter.github.io/nullius](https://bpachter.github.io/nullius)

Structural-biology evidence viewer. Published protein structures render every atom as equally certain, but the experiment measures an electron-density cloud and a person decides where the atoms go. Shows atom by atom which parts of a molecule the measurement supports, worst-evidence-first, from a precomputed 1.9M-row index of the entire PDB — one request, ~50 ms, no backend.

Stack — TypeScript · BinaryCIF decoder · WebGL point clouds · RDKit WASM · AlphaFold

## Interactive explainers & models

[bpachter.github.io](https://bpachter.github.io)

Gradient (how machines learn, 1958 perceptron to modern inference), Entropy (thermodynamics on a hand-written hard-sphere gas engine), The Great Attractor (cosmology, 18 live figures), Kernel (CPython on WebAssembly in-browser), Drawdown (EIA-calibrated US gas balance), Updraft (waste-heat atlas), and a Graph Ontology guide grounded in IEC CIM. Every simulation hand-written; no charting libraries.

Stack — TypeScript · React · Canvas 2D · WebGL / three.js · Pyodide · MapLibre GL

## 05 EDUCATION

### University of North Carolina at Charlotte

Charlotte, NC

BSBA — Operations & Supply Chain Management and Management Information Systems

## 06 CERTIFICATIONS

### IBM AI Engineering Professional Certificate

LLMs, Transformers, LoRA / QLoRA, RLHF, RAG · 13 courses

### Power BI PL-300 Data Analyst Associate

ETL, DAX, data modeling, Microsoft Fabric

### SAS Statistical Business Analyst

Regression modeling, hypothesis testing

### DeepLearning.AI / Stanford — ML Specialization

Supervised & unsupervised learning, algorithms, RL

### Architecting Solutions on AWS

Lambda, Glue, S3, Athena, Redshift, SageMaker